

Akshansh Ranjan

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EDUCATION

BITS Pilani, Hyderabad Campus

2023 – 2027

B.E. Electrical and Electronics Engineering, Minor in Computer Science

CGPA: 8.02

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Operating Systems, AI, Machine Learning

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, SQL, Java

Backend: Node.js, Express.js, FastAPI, PostgreSQL, Prisma, Redis, WebSocket, Socket.io, BullMQ

AI / LLM: LangGraph, LangChain, RAG, MCP, ChromaDB, OpenAI API, Gemini API, Hugging Face

Frontend: Next.js 14, React, Redux Toolkit, Tailwind CSS, Radix UI

DevOps / Tools: Docker, Docker Compose, GitHub Actions, Vercel, Git

EXPERIENCE

ML Intern

May 2025 – Jul 2025

AI Gurukul

Remote

- Fine-tuned **XLM-Roberta** for sentence-level classification on code-mixed text across **5 Indian languages**.
- Achieved **91% weighted F1-score** on held-out test data across all language classification tasks.
- Built a token-level language detector using **Gemini API + BM25** retrieval across **70+ languages**.
- Developed a PDF-based **RAG chatbot** with custom chunking, semantic retrieval, and multilingual prompting.
- Designed full **NLP data pipeline**: preprocessing, tokenization, label balancing, and F1/confusion matrix evaluation.

PROJECTS

OrqFlow – Enterprise Distributed AI Orchestration System | *FastAPI, Python, Redis, Postgres, Docker* Jun 2026

- Architected an async **11-endpoint FastAPI backend** utilizing Server-Sent Events (**SSE**) for low-latency data streaming.
- Engineered zero-downtime **API failover** switching seamlessly between **Groq** and **Gemini** upon HTTP 429 rate limits.
- Decoupled system tools into **3 FastMCP microservices** handling database queries, file manipulation, and web search.
- Containerized the stack via **Docker Compose** and validated core reliability with an **86-test regression suite** (<1.5s).
- Built Sliding-window **Redis rate-limiting** (10 req/min) and async **PostgreSQL schemas** with Alembic migrations.

SyncNexus – Real-time Collaboration with RAG AI Assistant | *Socket.IO, ChromaDB, BullMQ, Redis* April 2026

- Integrated **RAG-powered AI assistant** answering queries in < 3s with exact **metadata citations** via ChromaDB.
- Built scalable **document ingestion pipeline** for PDFs: S3 upload → 1k-token chunking → idempotent ChromaDB upserts.
- Used 768-dimensional **Gemini text embeddings** for semantic storage, **retrieving top-5 relevant chunks** at query time.
- Implemented async LLM processing via **BullMQ** with 3-attempt exponential backoff, preventing WebSocket blocking.
- Designed 5-minute TTL **presigned S3 uploads** and $O(\log N)$ **cursor pagination** with sub-ms Redis presence tracking.

LinkForge – URL Shortener & Analytics Engine | *Express, TypeScript, PostgreSQL, Prisma, Redis, Docker* Feb 2026

- Built production-grade URL shortener with **Redis cache-aside pattern**, cutting redirect latency from 45ms to 8ms.
- Implemented **sliding-window rate limiter** using Redis sorted sets — $O(\log n)$ per request per user/IP tier.
- Engineered click analytics pipeline tracking **geography, device, referrer, and temporal patterns** async.
- Containerized with **Docker Compose** (app + PostgreSQL + Redis) and deployed on Railway.
- Set up **GitHub Actions CI/CD** pipeline for automated testing and zero-downtime production deployments.

Axiom Pulse – Token Discovery Dashboard | *Next.js 14, TypeScript, NextAuth.js, OAuth 2.0, Radix UI* Nov 2025

- Architected a pixel-perfect crypto dashboard with **Next.js 14** and enterprise SSO via **NextAuth.js**.
- Implemented **Google + GitHub OAuth 2.0** authentication with session management and protected routes.
- Integrated **WebSocket mocks** for live token price updates with seamless Tailwind CSS transitions.
- Achieved **Lighthouse >90 (Desktop)** via skeleton loading states and component memoization.
- Built atomic **Radix UI** component architecture — all interactions under **100ms latency**, deployed on Vercel.

ACHIEVEMENTS

Published a research paper at IEEE GCON, IIT Guwahati introducing **Robust Difference Plot (RDP) Analysis**, a novel methodology for evaluating **Frequency Response Analysis (FRA)** plots to detect six different mechanical and electrical transformer faults. Jun 2026

National Finalist (Top 2.5%) – Meta PyTorch OpenEnv Hackathon × Scaler: Built **BluffBuster** using **OpenEnv**, **Unsloth**, and **GRPO** to train language models via multi-style student simulators and calibrated belief tracking. Selected among **800 teams from 31,000+ registrations**. Apr 2026

Extra-curricular: Represented school and university **football** teams in competitive tournaments.

Communication : Fluent in English and Hindi.